

Why isn't it enough to know
linear control theory?

Learning goals and Literature

After this set of video lectures you should...

- Know the basic differences between linear and nonlinear systems
- Recognize the need for new analysis and control design methods
- Know when to use nonlinear methods for analysis and design
- Know how to calculate equilibrium points

The video lectures are based on

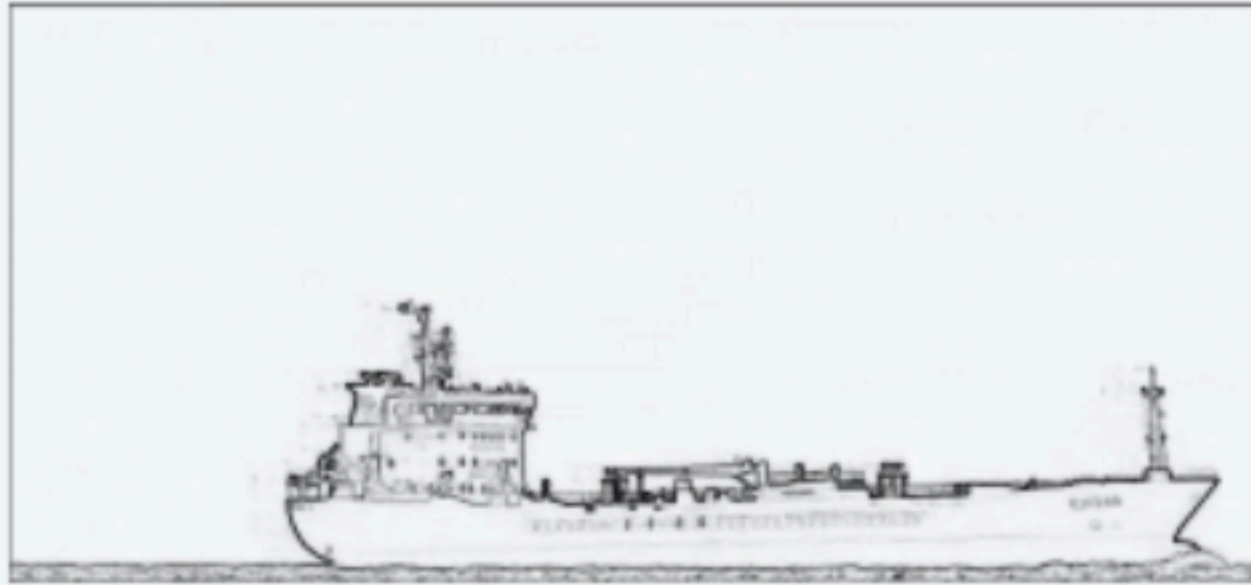
Khalil Chapter 1

Nonlinear dynamic systems



Process / System / Mechanism

Nonlinear dynamic systems



$$\dot{u} = \frac{m_{22}}{m_{11}} vr - \frac{d_{11}}{m_{11}} u + \frac{1}{m_{11}} u_1$$

$$\dot{v} = -\frac{m_{11}}{m_{22}} ur - \frac{d_{22}}{m_{22}} v$$

$$\dot{r} = \frac{m_{11} - m_{22}}{m_{33}} uv - \frac{d_{33}}{m_{33}} r + \frac{1}{m_{33}} u_2$$

$$\dot{x} = u \cos \psi - v \sin \psi$$

$$\dot{y} = u \sin \psi + v \cos \psi$$

$$\dot{\psi} = r$$

Mathematical Model

$$m \frac{d^2 x(t)}{dt^2} = F(x(t))$$

ODE

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

PDE

Nonlinear dynamic systems

Linear systems (LTI)

$$\dot{x} = ax + bu$$

$$u = -K_p(x + \frac{1}{T_i} \int x d\tau + T_d \dot{x})$$

$$z \triangleq [x, \dot{x}]^T$$

↓

$$\dot{z} = Az$$

Solution:

$$z(t) = e^{At}z(0)$$

Analysis:

Laplace transform →
Transfer function

Nonlinear systems

$$\dot{x} = f_p(t, x, u)$$

$$u = \gamma(t, x)$$

↓

$$\dot{x} = f(t, x)$$

Solution:

Generally no analytical solution

Analysis:

No Laplace transform
⇒ need new methods

Nonlinear dynamic systems

We consider nonlinear dynamic systems in the form

$$\begin{aligned}\dot{x} &= f_p(t, x, u) \\ \dot{x}_1 &= f_{p_1}(t, x_1, \dots, x_n, u_1, \dots, u_m) \\ &\vdots \\ \dot{x}_n &= f_{p_n}(t, x_1, \dots, x_n, u_1, \dots, u_m) \\ &\downarrow u = \gamma(t, x) \\ \dot{x} &= f(t, x) \quad \text{Time-varying / Nonautonomous}\end{aligned}$$

Special case

$$\dot{x} = f(x) \quad \text{Time-invariant / Autonomous}$$

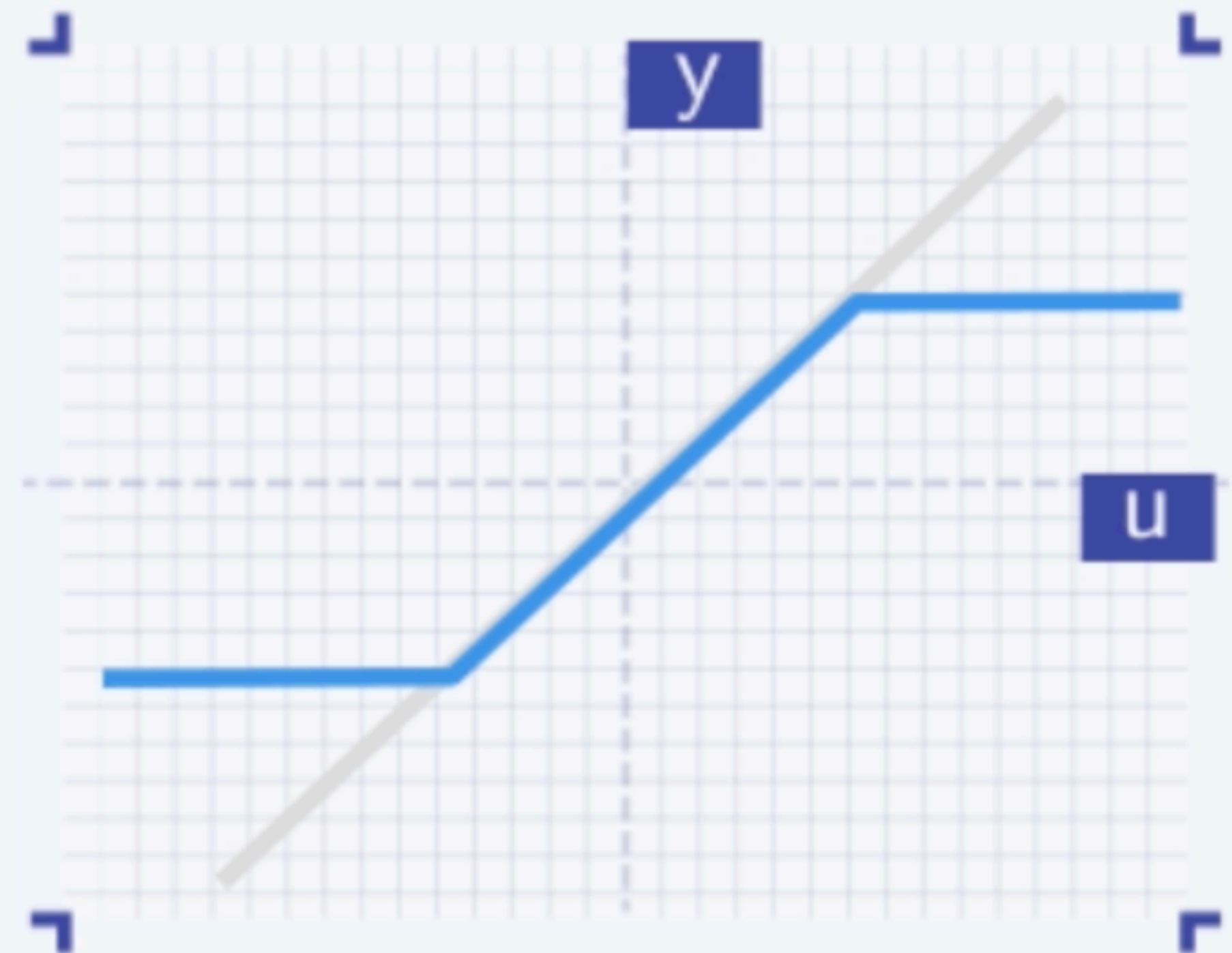
Examples of nonlinear systems

Systems with input saturation

$$\dot{x} = Ax + B\text{sat}(u)$$

$$u = \text{PID}(x)$$

$$\text{sat}(u) = \begin{cases} u & \text{if } |u| \leq 1 \\ \text{sgn}(u) & \text{if } |u| > 1 \end{cases}$$



Examples of nonlinear systems

Systems with essential nonlinearities in the model

- Electric motors in hybrid cars

$$\dot{\lambda}_r = \left(-\frac{R_r}{L_r} I + n_p \omega J \right) \lambda_r + \frac{R_r}{L_r} M i_s$$

$$\dot{i}_s = -\frac{M}{\sigma L_s L_r} \left(-\frac{R_r}{L_r} I + n_p \omega J \right) \lambda_r - \frac{1}{\sigma L_s} \left(R_s + \frac{M^2 R_r}{L_r^2} \right) i_s$$

$$\dot{\Theta} = \omega$$

$$\dot{\omega} = \mu i_s^T J \lambda_r - \frac{\alpha \sigma}{m} + \frac{T_L}{m}$$



Examples of nonlinear systems

- Systems with highly complex dynamics
 - Weather prediction
 - Lorenz attractor

$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = rx - y - xz$$

$$\dot{z} = xy - bz$$

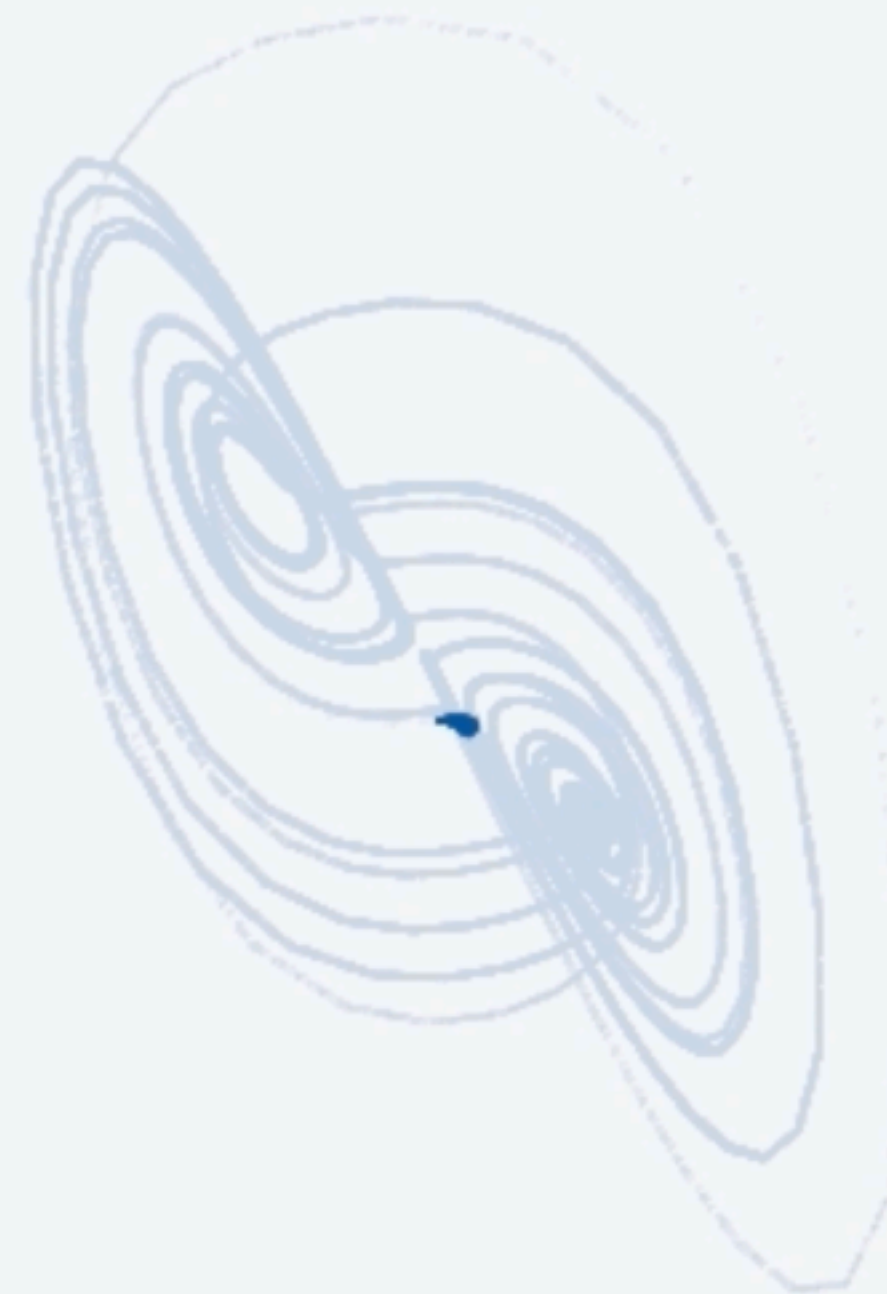


Figure : Lorenz attractor

Examples of nonlinear systems

- Systems "pushing the limit"
 - Systems operating in extreme conditions (where nonlinear effects become significant)



Examples of nonlinear systems

- Systems with complex non-local dynamics
 - HIV virus dynamics

$$\dot{T} = s - dT - \beta vT$$

$$\dot{T}_1 = q_1 \beta vT - \mu_1 T_1 - k_1 T_1$$

$$\dot{T}_2 = q_2 \beta vT + k_1 T_1 - \mu_2 T_2$$

$$\dot{v} = k_2 T_2 - cv$$

T number of healthy immune cells

T_1 number of infected, but non-reproducing cells

T_2 number of infected and virus reproducing cells

v number of free virus particles

The Superposition Principle

- For linear ODEs

Formulation 1

The response to several inputs working together is equal to the sum of the response of that circuit to the inputs working separately

Formulation 2

If input u_1 produces output y_1 , and input u_2 produces output y_2 , then input $u = u_1 + u_2$ produces output $y = y_1 + y_2$.

- For nonlinear systems

No superposition principle $\Rightarrow$ More complex behavior

The Superposition Principle

Leo Tolstoy's Anna Karenina

*All happy families are alike;
each unhappy family is unhappy in its own way.*

The Superposition Principle

Leo Tolstoy's Anna Karenina

*All happy families are alike;
each unhappy family is unhappy in its own way.*

Slight adaptation

*All linear systems are alike;
each nonlinear system is nonlinear in its own way.*

Invariant sets

We study the system's behavior in and around invariant sets

Equilibrium point

x^* is an equilibrium point of $\dot{x} = f(x)$ iff

$$f(x^*) \equiv 0$$

An equilibrium point is a special case of

Invariant sets

A set M is an invariant set of $\dot{x} = f(x)$ iff

$$x(0) \in M \quad \Rightarrow \quad x(t) \in M, \quad \forall t \in \mathbb{R}$$

Linear systems

- One equilibrium point

$$\dot{x} = Ax = 0$$

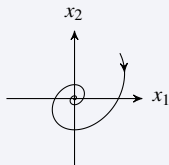


$$x^* = 0$$

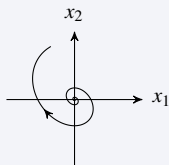
- May in addition have periodic solutions

$$x(t+T) = x(t)$$

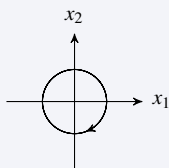
Invariant sets: Linear systems



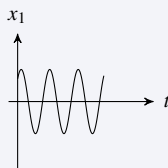
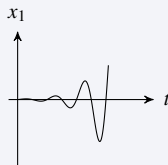
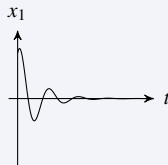
$x^* = 0$ as. stable



$x^* = 0$ unstable

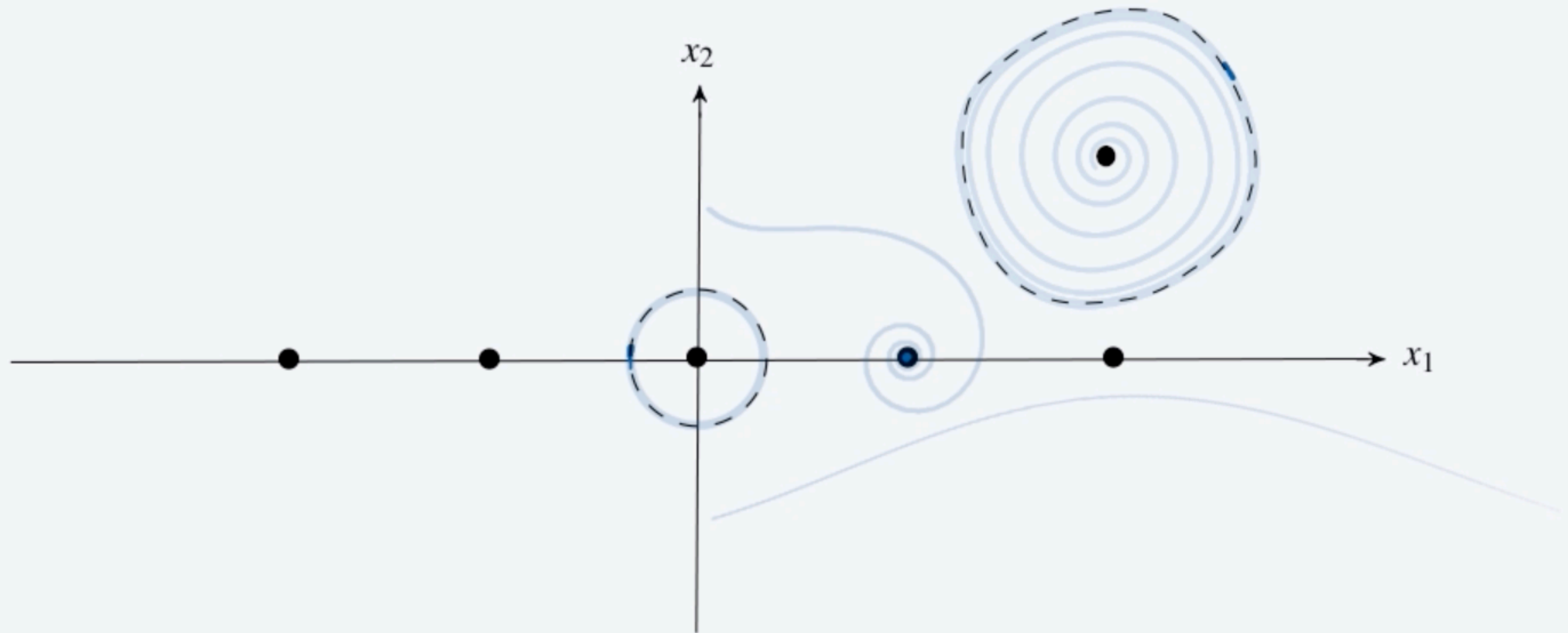


$x^* = 0$ marginally
stable



Invariant sets: Nonlinear systems

- Equilibrium points, one or more
- Periodic solutions
 - Limit cycles
- General invariant sets (Chaos)

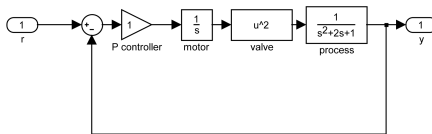


Common nonlinear phenomena

Stability may depend on the magnitude of r

- $\neq$ Linear systems: Stability properties local $\sim$ global

Example: Motor valve process

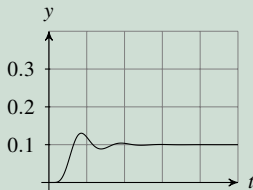


- Valve characteristic: $f(\theta) = \theta^2$
- Reference signals of different magnitude
 - $r = 0.1$
 - $r = 1$
 - $r = 1.72$

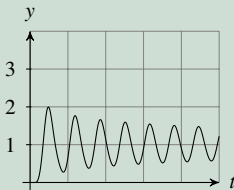
Common nonlinear phenomena

Step responses ($x_0 = 0$)

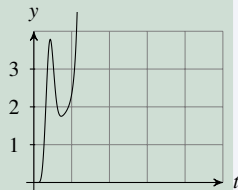
$r = 0.1$



$r = 1.0$



$r = 1.72$



Stability depends on the magnitude of r !

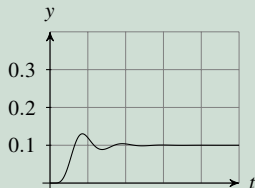
Common nonlinear phenomena

Stability may depend on initial conditions

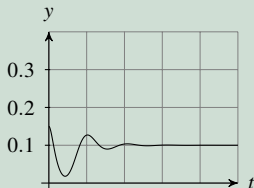
$$x_0 = x(t_0)$$

Step responses $r = 0.1$, for different x_0

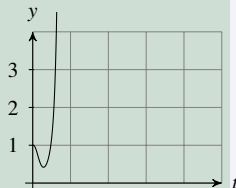
$x_0 = 0$



$x_0 = 0.15$



$x_0 = 1.0$



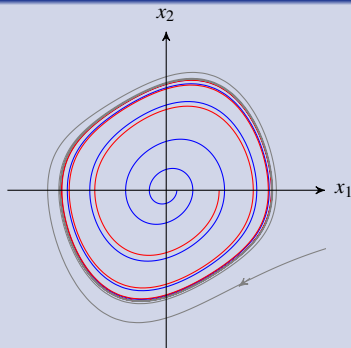
Common nonlinear phenomena

Stable periodic solutions (Limit cycles)

- No external periodic input
- Frequency and amplitude independent of initial state

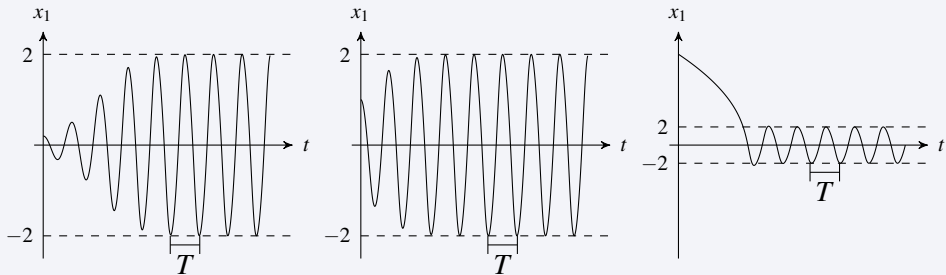
Example: Van der Pol

$$\ddot{x} + \varepsilon(x^2 - 1)\dot{x} + x = 0$$



Common nonlinear phenomena

Simulation results:



Frequency and amplitude independent on initial conditions!

Common nonlinear phenomena

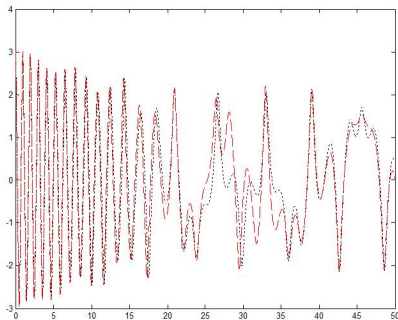
Chaos and Bifurcations

Example

$$\ddot{x} + 0.1\dot{x} + x^5 = 6\sin t$$

● - - - $x(0) = 2, \quad \dot{x}(0) = 3$

● - - - $x(0) = 2.01, \quad \dot{x}(0) = 3.01$



When to use nonlinear analysis/design

When should we use nonlinear analysis/design?

- Nontrivial question
- When you recognize essential nonlinear phenomena (that cannot be properly described by a linear model)
- When the workspace is large
- When the nonlinearities are hard

The nonlinear phenomena are dominant

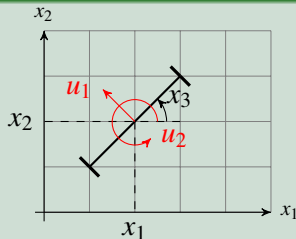
Example: Car

Kinematic model:

$$\dot{x}_1 = \sin(x_3)u_1$$

$$\dot{x}_2 = \cos(x_3)u_1$$

$$\dot{x}_3 = u_2$$



- Control problem: Parallel parking

Find $u = [u_1, u_2]^T$ that makes $x = [0, 0, 0]^T$ as. stable.

- Linearized model

$$\dot{x}_1 = 0$$

$$\dot{x}_2 = u_1$$

$$\dot{x}_3 = u_2$$

Not controllable

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